# KL Divergence

Math Foundations — Concept 38

## 1 Definition

Let $p$ and $q$ be probability distributions on a common measurable space $(\mathcal{X}, \mathcal{F})$, with $p$ absolutely continuous with respect to $q$ (written $p \ll q$). The *Kullback–Leibler divergence* of $p$ from $q$ is

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \mathbb{E}_p\!\left[\log \frac{dp}{dq}\right].$$

**Definition 1** (Discrete KL divergence)**.** *For probability mass functions $p, q$ on a countable set $\mathcal{X}$,*

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)},$$

*with conventions $0 \log 0 = 0$ and $0 \log(0/0) = 0$. The sum is $+\infty$ if there exists $x$ with $p(x) > 0$ and $q(x) = 0$.*

**Definition 2** (Continuous KL divergence)**.** *For probability densities $f_p, f_q$ on $\mathbb{R}^d$ (with respect to Lebesgue measure),*

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \int_{\mathbb{R}^d} f_p(x) \log \frac{f_p(x)}{f_q(x)}\, dx,$$

*defined whenever the integrand is integrable and $\{f_q = 0\} \subseteq \{f_p = 0\}$ up to null sets.*

**Remark 3.** *$D_{\mathrm{KL}}$ is* not *symmetric and does* not *satisfy the triangle inequality, so it is not a metric. Nonetheless it is non-negative, vanishes only when $p = q$, and behaves additively under independence, justifying its role as a "relative information" measure.*

## 2 Gibbs' inequality

**Theorem 4** (Non-negativity of KL)**.** *For any pair of probability distributions $p, q$ on the same space with $p \ll q$,*

$$D_{\mathrm{KL}}(p \,\|\, q) \;\geq\; 0,$$

*with equality if and only if $p = q$ almost everywhere.*

*Proof.* Write $r(x) = q(x)/p(x)$ on the support of $p$. By definition,

$$D_{\mathrm{KL}}(p \,\|\, q) \;=\; \mathbb{E}_p\!\left[\log \frac{p(X)}{q(X)}\right] \;=\; \mathbb{E}_p[-\log r(X)]\,.$$

The function $\varphi(t) = -\log t$ is strictly convex on $(0, \infty)$. By Jensen's inequality applied to $\varphi$ and the random variable $r(X)$ under $p$,

$$\mathbb{E}_p[-\log r(X)] \;\geq\; -\log \mathbb{E}_p[r(X)]\,.$$

We compute the right-hand side:

$$\mathbb{E}_p[r(X)] \;=\; \int r(x)\,p(x)\,dx \;=\; \int \frac{q(x)}{p(x)}\,p(x)\,dx \;=\; \int q(x)\,dx \;=\; 1,$$

since $q$ is a probability distribution. Therefore

$$D_{\mathrm{KL}}(p\,\|\,q) \;\geq\; -\log 1 \;=\; 0.$$

Because $\varphi$ is *strictly* convex, Jensen's inequality is an equality iff $r(X)$ is $p$-almost-surely constant. Combined with $\mathbb{E}_p[r(X)] = 1$, this forces $r \equiv 1$ on the support of $p$, i.e. $p = q$ a.e. The discrete case is identical with integrals replaced by sums. $\square$

**Remark 5.** *Run with $\varphi = -\log$ replaced by an arbitrary convex $\varphi$ with $\varphi(1) = 0$, the same proof yields the family of $f$-*divergences *(Csiszár). KL corresponds to $\varphi(t) = t\log t$ (after a sign convention); other choices recover total variation, Hellinger, and $\chi^2$ divergences.*

# 3 Consequences

**Lemma 6** (Cross-entropy decomposition)**.** *For all $p, q$ with $p \ll q$,*

$$H(p,q) \;=\; H(p) + D_{\mathrm{KL}}(p\,\|\,q),$$

*where $H(p) = -\mathbb{E}_p[\log p]$ and $H(p,q) = -\mathbb{E}_p[\log q]$.*

By Theorem 4, $H(p,q) \geq H(p)$ with equality iff $q = p$. Hence minimizing cross-entropy in $q$ for fixed $p$ is the same as minimizing KL divergence — the foundational identity behind maximum likelihood estimation and the categorical cross-entropy loss.